

Country Intelligence & Pipeline Engine

Phase 1 scope: the daily research and opportunity engine, built to full quality with no shortcuts. Phases 2–5 are trimmed to a one-page menu to pick from later, not a commitment.

An always-on research layer that reads a country's government, corporate and bilateral activity, qualifies the companies it finds against a revenue/scale bar, and surfaces up to 10 ranked, sourced, actionable opportunities a day — each with a stated reason a Director can act on immediately. This is the brief's own Phase 1, undiluted: everything it needs (scheduler, qualification engine, prioritisation, archive, suppression, cost governance) is specified to full acceptance-criteria depth below. HubSpot sync, the broader feed, watchlists, on-demand deep search and analytics are real, valuable, and deliberately **not** committed to yet — see §5 for what to pick from when it's time.

STORIES (NOW)	ROLES TOUCHED	EST. EFFORT (NOW)	MAPPED FOR LATER
17 · US-062–076, 095–096	user, admin	75 hours	18 stories, §5

0. SCOPE & SEQUENCING

This build is Phase 1 only — the brief's own daily engine — plus the foundation it sits on and two cross-cutting stories (privacy notice, project-scoped visibility) that apply the moment any user touches the module. Nothing in this scope is a stub or a simplified version of the real thing: the qualification engine, the six-dimension prioritisation logic, the resumable scheduler and the cost governance around unattended runs are all specified at full depth. What's deliberately left out is everything downstream of Phase 1 — HubSpot sync, the wider article feed, watchlists, on-demand search, analytics — because each of those is either blocked on something only TRC can provide (HubSpot credentials, field mapping) or is genuinely optional add-on value that's cheaper to scope precisely once Phase 1 is live and we know what a real project actually needs. §5 keeps all of that mapped at the same story-ID numbering as before, so nothing has to be re-specified from scratch later — it's a menu, not a rewrite.

What's genuinely new here vs. every existing module: every other AI workflow in this app is synchronous — a user clicks, Claude runs, output streams back, the human is in the loop for every dollar spent. This is the first **autonomous, unattended, recurring** workflow: it runs daily whether or not anyone opens the app. That single fact is why this scope carries dedicated stories for cost governance and resumable multi-pass runs that no other module needed.

1. ROLES

ROLE	ACCESS
user	Existing generic role, gated by a new <code>can_access_country_intelligence</code> flag — same pattern as every other module. Sees only the active projects they are assigned to (never an unrestricted country search). Reads opportunities, rates and saves items, searches the archive.
user	+ <i>Project Director</i> — not a new global role. A per-project elevation stored on a membership row (<code>country_intel_project_members.project_role IN ('director', 'member')</code>), the exact shape already used for <code>finance_project_members.project_role IN ('director', 'sales_rep')</code> in the Cash Box module. A Director additionally: reverses a "not relevant" suppression, reviews project activity. (Assignment/HubSpot-specific Director powers from the fuller brief apply once that phase is picked from §5.)
admin	Unrestricted, as everywhere else in the app. Creates/activates/pauses/closes projects, assigns Directors and teams, sees every project regardless of assignment, sets cost ceilings, reviews usage/cost.

Why a per-project role instead of a third global role: the brief's "Project team member vs. Project Director" split only matters inside this one module — a Director on one project is an ordinary team member on another. Modeling it as a project-scoped membership row (mirroring Cash Box) avoids adding a third value to the app-wide `UserRole` type, which every other module's access logic assumes is binary (admin/user). The same table carries every later phase's Director-only actions (HubSpot add, watchlist management, on-demand search) without a schema change when they're picked up.

KEY TECHNICAL DECISIONS

Scheduler. Vercel Cron triggers one dispatcher endpoint daily; it fans out one resumable job per active project rather than doing all research inline in the cron request itself. Each per-project run reuses the document module's **soft-deadline + incremental persistence + auto-continue** pattern (already built for segmented generation) instead of assuming a single request can finish a full daily research pass within the platform's request-time ceiling.

Structured, code-validated output for anything that scores or classifies. Prioritisation (US-071) and company qualification (US-070) use Claude's `json_schema` structured output plus a code-side validation/corrective-reprompt pass — the exact pattern already proven in the Sales Coach Report Card (`validateReportCard()`, one corrective retry, code owns the arithmetic). Freeform prose scoring was rejected there for the same reason it would fail here: an unvalidated model can quietly invent a number.

Everything past Phase 1 is deliberately unscheduled, not cancelled. The data model (US-064) is already shaped so HubSpot IDs, watchlist entries and feed items attach without a migration rewrite when one of those gets picked up — see §5.

2. USER STORIES — SUMMARY (THIS BUILD)

ID	AREA	STORY
FOUNDATION — DATA MODEL, ROLES, PROJECT LIFECYCLE		
US-062	● ACCESS	Module and Director elevation are gated per project, not by a new global role
US-063	● ADMIN	Admin creates, activates, pauses and closes country projects
US-064	● DATA	System stores articles, companies, people and opportunities as separate, connected records
PHASE 1 — DAILY PRIORITY OPPORTUNITIES		
US-065	● SCHEDULER	Activation starts a 3-week lookback, then a resumable daily refresh
US-066	● RESEARCH	System infers relevant sectors from the country's economic structure
US-067	● RESEARCH	System monitors government, presidential and ministerial activity
US-068	● RESEARCH	System monitors companies, leaders and corporate developments
US-069	● LOGIC	Bilateral developments between project and publication country get a relevance boost
US-070	● QUALIFICATION	Every company is qualified against the revenue/scale bar, never with invented numbers
US-071	● LOGIC	AI prioritisation scores and explains every opportunity across six dimensions
US-072	● OUTPUT	Up to 10 priority opportunities per day, each a full actionable card
US-073	● ACTIONS	User rates, saves or suppresses an opportunity; suppression sticks
US-074	● ARCHIVE	Every opportunity is preserved in a searchable per-project archive
US-075	● SOURCES	Every fact carries a source, date and an evidence-type label
US-076	● COST	Unattended daily runs are hard-capped and fully logged to the usage ledger
CROSS-CUTTING		
US-095	● PRIVACY	First-time users see and dismiss the country-intelligence data notice
US-096	● EDGE	A user only ever sees projects they are assigned to, never an unrestricted search

3. STORY DETAIL & ACCEPTANCE CRITERIA

§0 Foundation

Data model, project lifecycle and role plumbing everything else depends on — including the phases picked up later.

US-062 ACCESS

As a user, my module access and any Project Director elevation are both explicitly granted, never assumed

- New flag `can_access_country_intelligence` on profiles/invitations, wired through `access.ts/middleware/Sidebar` exactly like every other gated module
- New `country_intel_project_members` table (`project_id`, `user_id`, `project_role`) mirrors `finance_project_members` exactly — one partial-unique index enforcing at most one Director per project, an `is_country_intel_director(project_id)` RLS helper mirroring `is_finance_project_member()`
- A user without the module flag cannot reach the module via direct URL; a user with the flag but no membership row for a given project cannot reach that project's pages via direct URL either
- Admins bypass both checks, exactly as in every other module

US-063 ADMIN

As an admin, I create and control the lifecycle of each country project

- Create form captures: country, publication/media, publication country or principal market, Project Director (single user, must already have the module flag), team members (multi-select, same pool), start date
- Status is one of Active / Paused / Closed; only Active projects run daily research (US-065)
- Pausing stops automated daily research immediately but keeps the archive intact; reactivating resumes daily refresh without repeating the 3-week lookback
- Closing stops daily monitoring and normal refresh entirely for this build (watchlist-on-close from the fuller brief applies once that phase is picked up — see §5)
- Team roster and Director are editable after creation; changing Director reassigns the single-Director slot, it does not stack

US-064 DATA

As the system, I keep articles, companies, people and opportunities as separate, connected records

- Core tables: `country_intel_articles`, `_companies`, `_people`, `_opportunities`, `_sources`, `_government_institutions`, `_events`, plus join tables so one article can reference many companies/people and one opportunity can be backed by many articles — this is the brief's own §17 requirement, and it is what makes suppression (US-073) possible at all
- Company records carry a normalised-name field (lowercased, punctuation-stripped, alias array) used for the dedup/suppression logic in US-070/073 — built once here, deliberately shaped so it's also what HubSpot matching and watchlists will key off if/when those are picked up from §5, with no schema rewrite
- Every record is project-scoped (RLS on `project_id`) and carries `created_at/source-run` metadata for the archive

This is the highest-leverage story in the whole build — getting the normalisation right now is far cheaper than retrofitting it later if HubSpot sync or watchlists get picked up from §5.

§1 Phase 1 — Daily Priority Opportunities

The brief's own stated Phase 1, in full: the unattended daily engine, qualification logic, prioritisation, and the archive/actions loop around it.

US-065 SCHEDULER

As the system, I run a 3-week lookback on activation, then refresh daily, resumable across requests

- One Vercel Cron job (daily) hits a dispatcher endpoint that enumerates Active projects and enqueues one run per project — it does not do the research inline itself
- Each per-project run reuses the document module's soft-deadline pattern (a `setTimeout`-driven abort, not an in-loop check — see CLAUDE.md's note on why an in-loop check misses quiet stretches like a web-search round-trip) plus incremental persistence, so a run that spans multiple platform-imposed request windows resumes rather than restarts
- First run per project after activation covers the previous three weeks and surfaces the newest/most time-sensitive items first; every run after that is a true incremental refresh — it does not re-surface material already seen unless something changed (feeds directly into US-073's suppression)
- A run's status (pending/running/complete/failed) is visible to the admin per project, mirroring `research_sessions.status`'s pattern in the existing Research module

US-066 RESEARCH

As the system, I infer which sectors matter for this country without the Director configuring any

- One admin-editable, versioned prompt (mirrors `meeting_prep_prompt`'s pattern) holds TRC's standard sector universe (banking, telecoms, infrastructure, energy, manufacturing, construction, tourism, agriculture, logistics, healthcare, retail, defence, etc. — brief §4) as a checklist the model weighs against the country's actual economic structure
- Output is a ranked sector emphasis for this project (e.g. mining central, tourism secondary), stored on the project and re-evaluated on each cron cycle rather than fixed forever at activation
- Sector inference has no on/off toggle exposed to the Director — matching the brief's explicit "should not need to configure sectors manually"

US-067 RESEARCH

As the system, I track presidents, ministers, elections and major events for their commercial and access value

- Reuses the app's existing server-side `web_search` tool integration (the same `web_search_20250305` pattern already used in the document modules) rather than new search infrastructure, with a capped `max_uses` per project per run to bound cost
- For every leader/minister item, the model must answer the brief's own four questions (§6.2): why it matters to the report, which sectors/companies are involved, which CEOs may become relevant, whether it creates an interview or access opportunity — these four fields are part of the structured output schema, not left to prose
- Election/political-transition coverage is explicitly scoped to investment and access implications (brief §6.4) — the prompt instructs the model to drop general political coverage that has no commercial or access angle

US-068 RESEARCH

As the system, I track companies, leaders and corporate developments for both narrative and person

- Every priority company gets both a company-narrative pass (brief §6.7) and a leader-trajectory pass (§6.8) — the leader must never be dropped in favour of company-only coverage, per the brief's explicit "the person must not be forgotten behind the company"
- Leader research draws only on public, professionally-relevant information (career, achievements, public statements, board seats) — the prompt explicitly excludes speculation about private life and states gender/ethnicity/cause representation are not selection criteria, matching brief §6.8 verbatim
- Rankings/lists sources (largest companies, top employers, sector leaders — brief §6.9) are a distinct research pass whose output retains the ranking's source and date, feeding company discovery for US-070

US-069 LOGIC

As the system, I boost anything that connects the project country and the publication country

- A fixed checklist from brief §5 (state visits, trade missions, bilateral investment/trade agreements, joint ventures, defence cooperation, diplomatic anniversaries, delegations including business leaders) is evaluated against every candidate item
- A match sets a boolean bilateral flag plus a short explanation ("why this connects the two countries") that is carried onto the opportunity card (US-072) — this is what lets an item that's only moderately relevant domestically still surface as a top-10 opportunity
- Publication country is read from the project record (US-063); changing a project's publication country does not retroactively re-score already-surfaced opportunities

US-070 QUALIFICATION

As the system, I qualify every company against the revenue/scale bar and never invent a number

- USD 60M annual revenue is the default qualification bar (brief §7.1); when revenue is found, currency, reporting year, USD conversion and source are all recorded alongside it
- When revenue is unavailable, the company is **not** excluded — it is assessed against the brief's scale-indicator checklist (§7.2: employees, market share, listed status, market cap, branch count, major contracts, group membership, etc.) and assigned one of four verdicts: `qualified` / `likely_qualified` / `scale_uncertain` / `not_qualified`
- Excluded categories (academics, NGOs, start-ups, small businesses, speculative ventures — §7.3) are filtered before qualification runs, not after
- Structured output + a code-side validator (the Sales Coach Report Card pattern) rejects any qualification verdict that lacks either a revenue figure with a source, or at least one cited scale indicator — a bare "qualified" with no evidence is a validation failure that triggers one corrective re-prompt, not a silent pass-through

Brief §21: "Revenue and company size must never be invented" is listed as a core product principle, not a nice-to-have — this is why qualification gets its own validation pass rather than trusting the model's JSON directly, exactly as the Report Card's `validateReportCard()` already does for a different kind of unverifiable-model-output risk.

US-071 LOGIC

As the system, I score and explain every candidate opportunity across six fixed dimensions

- Editorial relevance, commercial potential, leader relevance, strategic access, bilateral relevance (US-069's output), timeliness and evidence quality (brief §15) are each scored via structured output, with a required one-line rationale per dimension
- The combined score determines both the daily top-10 cut (US-072) and the recommended action category (immediate outreach / strengthen existing approach / seek introduction / request government support / research further / monitor — brief §10.3)
- The rationale is always shown to the user alongside the opportunity — brief §15's "Users should not receive an unexplained score" is a hard requirement, not a stretch goal

US-072 OUTPUT

As a user, I see up to 10 priority opportunities a day, each a complete, actionable card

- Ranking follows the brief's stated priority order (§10.1): qualified CEOs/leaders first, then companies needing the right contact identified, then presidents/PMs with exceptional access value, then ministers, then other institutions — commercial company/CEO opportunities form the majority of the daily selection by construction, not by accident
- Card shows every field from brief §10.3: type, company/institution, leader, position, sector, development, company narrative, leader trajectory, editorial and commercial rationale, revenue/scale verdict, government/event/bilateral connections, sources, recommended action
- One opportunity may be backed by several related articles (US-064's join table) — the daily selection consolidates rather than surfacing near-duplicate items

US-073 ACTIONS

As a user, I rate, save or suppress an opportunity, and suppression actually sticks

- Four actions per item: Useful, Not useful, Save, Not relevant — recorded per user and per project
- "Not relevant" writes a suppression row keyed to the company's normalised-name (US-064); every subsequent daily run checks this table before surfacing that company again for this project
- Only a Project Director can reverse a not-relevant decision (brief §10.4) — a regular team member's not-relevant action is final from their own side but reversible by the Director

US-074 ARCHIVE

As a user, I can search every opportunity this project has ever surfaced

- Full-text + filtered search (company, person, sector, date, action category) over every historical opportunity, not just the current day's top 10
- Archive entries retain their original prioritisation score and rationale as of the day they were surfaced — a later prompt edit never rewrites archived history (same "snapshot, don't re-derive" principle already used for prompt versioning elsewhere in the app)

US-075 SOURCES

As a user, I can see exactly where every fact came from and how confident it is

- Every factual item carries: source, direct link, publication date, headline, language, short summary, and a confidence flag for incomplete information — brief §9's full list
- Each fact is labelled as one of: verified fact / official announcement / company claim / media report / AI inference / unconfirmed — this label is part of the structured output, not inferred by the UI after the fact
- Research prompts are instructed to prioritise government/presidential/ministry sites, company filings, stock-exchange disclosures, reputable business media and recognised financial databases over low-quality aggregators or uncorroborated social posts (brief §9); where multiple articles cover the same development, the system consolidates them and keeps the strongest sources rather than surfacing near-duplicates

US-076

COST

As an admin, unattended daily research is hard-capped and I can see exactly what it's spending

- Every project has an admin-set daily search/token ceiling (mirrors the reserve-based gating pattern used everywhere else in the app, e.g. MEETING_PREP_*_RESERVE); a run that would exceed it stops cleanly rather than overrunning
- A dedicated country_intel_daily_research workflow (plus one per research pass: government, corporate, qualification, prioritisation) is logged to the existing usage_events ledger, so this recurring, no-human-in-the-loop spend shows up in /admin/analytics exactly like every on-demand workflow already does
- Because nobody is clicking "Generate" to consciously absorb the cost here — the one real behavioural difference from every other module in this app — admin gets a per-project daily-spend alert threshold and a project-level pause-on-overflow option, not just after-the-fact visibility

CLAUDE.md's existing rule "Token limit check happens before API call — never after" was written for on-demand, human-triggered calls. An unattended daily cron has no human to block in the moment, so the equivalent control here is a hard per-project ceiling checked before each research pass starts, plus the ledger entry regardless of outcome — this story is the daily-engine's version of that same principle.

§X Cross-cutting

Apply the moment any user opens the module, independent of which later phases get picked up.

US-095

PRIVACY

As a first-time user, I see the data-use notice once and can find it again later

- Brief §19's exact notice text, shown once per user on first entry to the module, dismissed with "Got it"
- Reuses the acknowledgement-timestamp pattern already built for Sales Coach (sales_coach_privacy_ack_at) — a new country_intel_privacy_ack_at column on profiles, same dismiss/never-reappear/discreet-link behaviour

US-096

EDGE

As a user, I can never reach or search a project I'm not assigned to

- Direct-URL access to a project the user has no membership row for is blocked (US-062's RLS), matching this app's existing "no-access-via-direct-URL" pattern used for every other gated module
- The module never exposes an unrestricted country search to a non-admin user — every research action here is always scoped to a project the user is a member of

4. EFFORT ESTIMATE — THIS BUILD

STORY	SCOPE	HOURS
FOUNDATION		
US-062	Project-level Director role (mirrors finance_project_members) + module gate	3
US-063	Admin project CRUD: create/activate/pause/close, Director + team assignment	4
US-064	Core connected schema: articles/companies/people/opportunities/sources + normalisation	8
PHASE 1 — DAILY PRIORITY OPPORTUNITIES		
US-065	Cron dispatcher + resumable per-project run (soft-deadline, incremental persistence)	10
US-066	Sector-inference prompt + versioning	3
US-067	Government/presidential/ministerial research pass	5
US-068	Corporate/leader research pass (narrative + trajectory)	5
US-069	Bilateral relevance boost logic	4
US-070	Qualification engine: revenue/scale tiers + validator + corrective retry	6
US-071	Six-dimension structured prioritisation + rationale	6
US-072	Daily top-10 output, ranking + opportunity card	5
US-073	Rate/save/suppress actions + suppression enforcement	3
US-074	Searchable opportunity archive	3
US-075	Source/evidence-type capture and display	3
US-076	Per-project daily cost caps + usage_events workflows + alerting	4
CROSS-CUTTING		
US-095	Privacy notice (reuses ack-at pattern)	1
US-096	Project-scoped visibility enforcement	2
Total — this build		75

5. LATER — A MENU, NOT A COMMITMENT

Everything below is real, specified in the original brief, and genuinely valuable — it's just not part of this build. IDs are reserved (continuing from where this scope leaves off) so nothing needs renumbering when a piece gets picked up; none of it is estimated into the 75-hour total above. Pick items individually rather than treating this as "Phase 2" you must take as a block — e.g. on-demand search (US-088–091) has no dependency on HubSpot and can be picked up on its own.

ID	AREA	STORY	HRS
HUBSPOT INTEGRATION		BLOCKED ON TRC CREDENTIALS	
US-077	● INTEGRATION	Admin connects and manages the HubSpot integration	3
US-078	● CRM	Every company/contact cross-checked against HubSpot with fuzzy matching	6
US-079	● CRM	Companies already in HubSpot keep receiving new intelligence against their record	2
US-080	● CRM	Director adds a company/contact with mandatory assignment + duplicate resolution	5
US-081	● CRM	Government contacts classified separately from commercial prospects	2
US-082	● CRM	System tracks what happened after an opportunity reached HubSpot	3
FULL INTELLIGENCE FEED & MONITORING			
US-083	● FEED	Up to 20 broader articles/day, separate from the top-10 opportunities	4
US-084	● FEED	Article card with why-it-matters + same action set as an opportunity	2
US-085	● ARCHIVE	Feed archive filterable by company, person, sector, source, date	3
US-086	● WATCHLIST	Director adds a company or leader to the project's shared watchlist	3
US-087	● WATCHLIST	Daily research flags meaningful new developments for watched entities	4
ON-DEMAND COMPANY & SECTOR RESEARCH		NO HUBSPOT DEPENDENCY	
US-088	● ONDEMAND	Director runs a broad, sector, company, CEO or portfolio search on demand	5
US-089	● ONDEMAND	Each result returns a full sourced company and leader profile	4
US-090	● ONDEMAND	Results can be filtered and refined without re-running research	3
US-091	● ONDEMAND	Director bulk-reviews and selectively approves results into HubSpot	3
ANALYTICS & OPTIMISATION			
US-092	● ANALYTICS	Opportunity-to-meeting outcomes by country, sector, source, signal	5
US-093	● ANALYTICS	Research cost, volume and quality per project in one dashboard	3
US-094	● NOTIFY	Exploratory delivery channels flagged for a later build (e.g. WhatsApp)	2
If all of the above were picked up			58

Recommended pick-up order, if/when: on-demand research (US-088–091) first — it's standalone, needs no cron and no HubSpot, and reuses the qualification/prioritisation engine this build already delivers, so it's mostly UI + a bounded request route. HubSpot (US-077–082) next, once TRC supplies credentials and a field map. Feed/watchlist (US-083–087) and analytics (US-092–094) last, since they compound in value the longer Phase 1 has been running and there's real archive data to watch trends in.

6. POSSIBILITIES — WHAT PHASE 1 ALONE UNLOCKS

1. **A real, defensible reason to call, every single day.** The brief's central complaint about generic news aggregation is correct and solvable: the six-dimension scoring plus the mandatory "why it matters / why now / who to contact" fields (US-071/072) turn a news item into an outreach script's first line, automatically — this is delivered in full by this build alone.
2. **The CEO stops disappearing behind the company.** Because US-068 forces a leader-trajectory pass on every priority company, TRC's interview prep gets materially better even on companies already in the pipeline — this pays off in editorial quality independent of HubSpot ever being connected.
3. **Bilateral framing is a genuine differentiator.** Nobody else pitching a Nigeria report to USA TODAY is going to open with "Nigeria–US relations, this week" — US-069 makes that TRC's default posture, not a manual research task a Director has to remember to do.
4. **The suppression loop compounds from day one.** Every "not relevant" makes the next day's run smarter for that specific project — value that increases the longer a project stays active, independent of any later phase.
5. **Nothing here is a dead end.** Because the data model (US-064) and qualification/prioritisation engine (US-070/071) are built once, properly, every item in §5 is materially cheaper to add later than it would be to build cold.

7. OPEN QUESTIONS FOR TRC — PHASE 1 ONLY

1. **Per-project daily budget ceiling.** The brief doesn't set one — US-076 needs an admin-configurable default so the first project doesn't run at an undefined cost. What's an acceptable daily AI spend per active project?
2. **Source allowlist / blocklist for non-English government sites.** Which local-language government and regulator sites should be treated as authoritative per priority market, and are there any sources TRC wants explicitly excluded?
3. **Cron schedule and timezone.** One fixed UTC time for all projects, or a per-country local-morning refresh? Affects how "daily" is defined for a report team working across time zones.
4. **Team size and project count expectations.** Roughly how many active projects and team members per project should this build be sized and cost-tested against?

HubSpot field mapping and a WhatsApp delivery decision only need answering once those items are picked up from §5 — not before this build starts.